

CHEMISTRY BOARD QUESTION PAPER : MARCH 2023**Time: 3 Hrs · Max Marks: 70****General Instructions:**

The paper has four sections.

Section A: Q1 (10 MCQs × 1 mark), Q2 (8 very short × 1 mark).

Section B: Q3–Q14 (12 questions × 2 marks; attempt any 8).

Section C: Q15–Q26 (12 questions × 3 marks; attempt any 8).

Section D: Q27–Q31 (5 questions × 4 marks; attempt any 3).

Log table allowed, calculator not allowed.

Write MCQ answer with alphabet (a)/(b)/(c)/(d). Only first attempt is checked.

Given: $R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$, $N_A = 6.022 \times 10^{23}$, $F = 96500 \text{ C}$.

SECTION – A**Q.1 MCQs:**

i. BCC relation:

(A) $3r=4a$ (B) $r=3a/4$ (C) $r=\sqrt{3} a/4$ (D) $r=\sqrt{2} a/4$

ii. $\text{pH} = 11.2$, $[\text{OH}^-] = ?$

(A) 1.585×10^{-3} (B) 3.010×10^{-11} (C) 3.010×10^{-3} (D) 1.585×10^{-11}

iii. First-order gas-phase rate law:

(A) $(2.303/t) \log(P_i/(P_i - P_t))$ (B) $(2.303/t) \log(P_i/(2P_i - P_t))$

(C) $(2.303/t) \log(2P_i/(P_i - P_t))$ (D) $(2.303/t) \log((P_i - P_t)/2P_i)$

iv. Spin-only μ of Mn^{2+} : (A) 4.901 (B) 5.916 (C) 3.873 (D) 2.846 BM

v. Tetraamminedibromoplatinum(IV) bromide:

(A) $[\text{PtBr}(\text{NH}_3)_4]\text{Br}_2$ (B) $[\text{PtBr}_2(\text{NH}_3)_4]\text{Br}$ (C) $[\text{PtBr}_2(\text{NH}_3)_4]\text{Br}_2$ (D) $[\text{PtBr}(\text{NH}_3)_4]\text{Br}$

vi. Allylic halide: (A) $\text{R}-\text{CHX}-\text{R}$ (B) $\text{CH}_2=\text{CH}-\text{X}$ (C) $—$ (D) $\text{CH}_2=\text{CH}-\text{CH}_2-\text{X}$

vii. $\text{CH}_3-\text{CH}=\text{CH}-\text{CH}_2-\text{CHO} + \text{LiAlH}_4/\text{H}_3\text{O}^+ \rightarrow ?$

(A) $\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{OH}$ (B) $\text{CH}_3-\text{CH}=\text{CH}-\text{CH}_2-\text{CH}_2-\text{OH}$

(C) $\text{CH}_3-\text{CH}_2-\text{CH}_2-\text{CH}_2-\text{COOH}$ (D) $\text{CH}_3-\text{CH}=\text{CH}-\text{CH}_2-\text{COOH}$



viii. Ozonolysis of 2,3-dimethylbut-2-ene $\rightarrow$

(A)acetaldehyde (B)propionaldehyde+acetone (C)acetone(D)acetaldehyde+butyraldehyde

ix. Maltose linkage: (A) α , β -1,2 (B) α -1,4 (C) β -1,4 (D) α -1,6

x. Monomer of natural rubber: (A)isoprene (B)acrylonitrile (C) ϵ -caprolactam
(D)tetrafluoroethylene

Q.2 Very Short Answers:

i. Name technique used to know nanoparticle geometry.

ii. Product when acetamide reacts with LiAlH_4 /ether.

iii. Product when chlorobenzene reacts with sodium/dry ether.

iv. Chemical composition of cryolite.

v. Name platinum complex used in cancer treatment.

vi. SI unit of cryoscopic constant.

vii. Condition for spontaneity (ΔG).

viii. Molar conductivity of 0.5M BaCl_2 ($\kappa = 0.01 \Omega^{-1} \text{cm}^{-1}$).

SECTION – B

(Attempt any 8) [16 Marks]

Q.3 Lanthanides vs Actinides.

Q.4 Mole fraction of solute: Vapour pressure pure = 640 mmHg, solution = 600 mmHg.

Q.5 Define green chemistry; write two advantages of nanoparticles.

Q.6 Substitutional impurity defect; Interstitial impurity defect.

Q.7 Reactions: (i) Chlorobenzene + fuming H_2SO_4 (ii) Ethyl bromide + AgOAc .

Q.8 Acidic buffer; solubility– K_{sp} relation for PbI_2 .

Q.9 Action on ethylamine: (i) $\text{CHCl}_3 + \text{KOH}$ (ii) HNO_2 .

Q.10 Standard ΔG for $\text{Cd} + \text{Sn}^{2+} \rightarrow \text{Cd}^{2+} + \text{Sn}$.

Q.11 Glucose from sucrose; structure of D-ribose.

Q.12 Define extensive property; work done by expansion in vacuum ($10 \rightarrow 20 \text{ dm}^3$).

Q.13 Reactions for formation of nylon-6,6.

Q.14 Structures of chloric acid and peroxydisulphuric acid.

SECTION – C

(Attempt any 8) [24 Marks]

Q.15 Define osmosis; determine molar mass from elevation in boiling point.

Q.16 Conversions: $\text{EtOH} \rightarrow$ ethyl acetate; phenol $\rightarrow$ benzene; ether $\rightarrow$ ethyl chloride.

For more study material, question papers & updates — Scan the QR code or visit



- Q.17 Weak acid 10% dissociated at 0.05M $\rightarrow$ % dissociation at 0.15M.
- Q.18 Dehydrohalogenation of 2-chlorobutane; uses & environmental effects of CFCs.
- Q.19 Reversible isothermal work: 20 $\rightarrow$ 30 L, n=2 mol, T=300K.
- Q.20 Interstitial compounds; classification of alloys with examples.
- Q.21 H₂-O₂ fuel cell diagram; two applications.
- Q.22 Formation of [CoF₆]³⁻: hybridisation, magnetic properties, inner/outer, geometry.
- Q.23 Pseudo 1st order; derive zero-order integrated rate law.
- Q.24 Aldol condensation of ethanal.
- Q.25 Anomalous behaviour of oxygen (atomicity, magnetic property, oxidation states).
- Q.26 Conversions: acetic acid $\rightarrow$ acetic anhydride; acetic acid $\rightarrow$ ethanol; IUPAC name & structure of methylphenylamine.

SECTION – D

(Attempt any 3) [12 Marks]

- Q.27 Prove $t_{99.9} = 3 \times t_{90}$; electronic configuration of Gd (Z=64); nanomaterial used in tyres.
- Q.28 Derive relation between ΔH & ΔU ; define vulcanization; what is peptide bond?
- Q.29 Density of Ag (fcc, a=400 pm, M=108); note on haloform reaction.
- Q.30 Define diastereoisomers; cis/trans of [Co(NH₃)₄Cl₂]⁺; reference electrode; why ozone bleach is dry bleach.
- Q.31 Dow process for phenol; action of bromine water on phenol; why Group-16 has lower ionisation enthalpy than Group-15; uses of dioxygen.

